Provenance-Aware Multimodal Evidence Aggregation in Multi-Agent Systems

CS 562 — Project Report

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October 28, 2026

1. Introduction and Problem Statement

Multi-agent systems combine reports from several agents before making a decision. Agreement can appear reassuring, but several reports may originate from one observation. An image may be forwarded, cropped, compressed, and described by different agents; a sensor reading may be copied into several summaries. Treating these reports as independent observations can amplify confidence without adding independent acquisitions. In a security-sensitive workflow, this can cause an action to be approved before the available evidence justifies it.

For example, an inspection team receiving an original frame, a resized copy, and a caption has one acquisition despite receiving three reports. Three separately sampled sensor observations can instead provide additional evidence under a specified conditional-independence model. Aggregation must preserve this distinction while allowing multiple interpretations to improve understanding of the original observation.

Problem statement. Report-level aggregation can count multiple representations of a single acquisition event as separate support. This creates a mismatch between the observations available to the system and the confidence used by its action gate. The technical problem is to preserve observation identity across transformations and agent communication, consolidate dependent reports before aggregation, and prevent duplicate-derived confidence from triggering premature approval.

The central research question is whether trusted evidence lineage reduces confidence inflation and premature approvals more effectively than prompt instructions or content matching, particularly when evidence changes representation or modality. A successful defense must also retain approvals supported by genuinely independent observations.

We propose a controlled benchmark, a minimal provenance interface, and a root-aware consolidation procedure. The implementation uses one vision-capable model, a small worker pipeline, and a simulated binary action gate. Physical robots, live sensors, and general-purpose authorization infrastructure remain outside the prototype.

2. Background and Related Work

2.1 Evidence Dependence and Epistemic Sybils

Bara's *Epistemic Sybil Resistance* distinguishes multiplying agents from multiplying evidence [1]. Reports may appear different while descending from the same observation, and independent observations may produce similar reports. Consequently, report similarity alone cannot generally identify evidential ancestry.

An important qualification is that shared-root reports need not be completely redundant: repeated extraction may improve interpretation of a fixed source. Our project therefore distinguishes additional information about an existing acquisition from additional independent acquisitions. Consolidation

should preserve useful interpretations without treating each interpretation as another measurement.

Gao et al. examine restated evidence in controlled logs, web documents, and agent-team communication [2]. Their source-counting oracle and copy-weight measurement provide a methodological starting point. We extend this controlled comparison to image transformations and image-to-text reports.

2.2 Provenance and the Contribution Boundary

Dependency provenance records how outputs depend on inputs [3]. In our setting, a derivation graph links artifacts and reports to acquisition events. A crop, caption, and forwarded report can retain a common ancestor even when their byte content differs.

This graph captures computational ancestry; it does not establish that a measurement is truthful or statistically independent of every other acquisition. Exact hashes identify identical bytes, while perceptual matching estimates visual resemblance. Neither generally establishes ancestry across modalities.

Our contribution is a bounded empirical extension: evaluating when explicit lineage improves multimodal evidence aggregation relative to instructions and content matching. We do not claim that evidence dependence or provenance-aware aggregation is new.

3. Proposed Methods

3.1 Threat Model and Controlled World

A trusted harness generates a balanced binary hidden state $Y \in \{0,1\}$ and observation events. Each controlled sensor event reports the correct state with reliability $q = 0.75$, with errors conditionally independent given Y. Copies inherit their original event's value and are never resampled.

The harness renders observations as simple visual cards and structured sensor records. This provides an exact reference posterior while introducing image perception, transformations, and cross-modal reporting. A separate rendered-scene condition examines less structured visual behavior without claiming a calibrated Bayesian oracle.

The attacker may replay, transform, and route an available artifact through several workers. The attacker cannot modify harness-maintained lineage or hidden-state labels. Core experiments first isolate repetition without adversarial instructions. An optional indirect prompt-injection stress test asks a worker to describe copied evidence as independent confirmation.

Forged acquisition roots, colluding capture devices, and undeclared environmental dependencies remain outside the trusted-lineage assumption.

3.2 Pipeline and Provenance Interface

The pipeline contains observation generation, artifact transformation, worker reporting, consolidation, and final aggregation. Three worker roles inspect assigned artifacts and produce structured claims. The aggregator returns a probability $\hat{p} = P(Y = 1)$ and cited evidence identifiers.

Each artifact carries:

`evidence_id`, `root_ids`, `parent_ids`,
`operation`, `modality`.

Roots identify acquisition events, not camera identities, agent names, or report identifiers. A reused

frame retains its root; a newly acquired observation receives a new root. Transformations inherit the union of their parents' roots.

The harness attaches and validates metadata outside model-generated report text. Workers cannot create acquisition identifiers or relabel a copied artifact as an independent event. Unknown identifiers, missing parents, and invalid derivations are marked as unresolved lineage.

3.3 Consolidation Procedure

The core benchmark uses single-root worker reports. Consolidation proceeds as follows:

1. **Validate lineage.** Check artifact identifiers and parent relationships against the harness registry. Reports with invalid or missing lineage enter an unknown-dependence group.
2. **Group by acquisition root.** Collect all reports derived from the same observation event. Agent identity and report count do not create additional roots.
3. **Reconcile interpretations.** If reports agree on a root's value, retain one claim and its supporting report identifiers. If they disagree, a fixed reconciliation prompt reviews the interpretations and canonical root artifact, returning one value or an unresolved flag. The reconciliation model receives no hidden-state label.
4. **Emit one record per acquisition.** Provide the final aggregator with one consolidated record per validated root. Preserve disagreement and uncertainty in that record, but do not present repeated agreement as additional independent measurements.
5. **Apply the decision policy.** The deterministic gate approves when $\hat{p} \geq \tau$, with $\tau = 0.90$. If parsing, lineage, or reconciliation remains unresolved, the core defense abstains.

Reconciliation calls are included in runtime and token costs. An abstention cannot count as a successful retained approval.

Multi-root summaries are excluded from the core benchmark. A later ablation links each summary to its component roots and prevents counting the summary alongside those roots as additional acquisitions. This avoids treating overlapping root sets as independent sources.

3.4 Experimental Conditions and Baselines

Condition	Construction and intended comparison
Single observation	One artifact and one report establish baseline confidence.
Exact replay	The same artifact is presented to three workers with different agent identities.
Transformed copies	An original, a resized/recompressed copy, and a crop preserving the relevant region share one acquisition root.
Cross-modal replay	An image, its caption, and a structured description derive from the same acquisition.
Distinct observations	Three independently sampled events provide the positive control for genuine corroboration.
Related views	Separate views of one scene provide an exploratory correlated-evidence condition; independence is not assumed.

Table 1: Conditions separating representation count, agent count, and acquisition count.

We compare five methods:

1. naive aggregation of all reports;
2. aggregation with an instruction that copies add no independent acquisitions;
3. exact byte-hash deduplication;
4. perceptual-image matching followed by consolidation;
5. trusted-lineage consolidation.

Content-matching methods use the same reconciliation procedure after forming groups, so comparisons primarily test how groups are identified. Matching thresholds are calibrated on pilot instances and frozen before the main evaluation. Exact and perceptual matching operate on available artifacts; unresolved cross-modal relationships are recorded explicitly.

3.5 Budget, Controls, and Ablations

A separate ten-instance pilot validates parsing, provenance propagation, crop readability, and costs. The main evaluation uses 100 base instances, six conditions, and five methods: approximately 3,000 aggregation evaluations before repeat and reconciliation calls. Worker outputs are cached and reused across methods.

Report order and agent labels are randomized using logged seeds. The same hidden states and source events are reused across conditions wherever possible. A fixed 20-instance subset receives three total evaluations per method to estimate output variability. These repeats do not increase the number of independent instances.

A metadata-removal ablation preserves report content and record formatting while withholding ancestry. A diagnostic shuffles root labels to test whether valid lineage, rather than metadata presence alone, explains performance. Crops are audited to ensure that information loss is not mistaken for an evidence-dependence effect.

3.6 Metrics and Statistical Analysis

For prior $P(Y=1)=1/2$, the controlled-task oracle counts each distinct acquisition once:

$$p_i^* = \sigma\left(\sum_{r\in R_i} x_r \log\frac{q}{1-q}\right), \qquad \sigma(z)=\frac{1}{1+e^{-z}}, \tag{1}$$

where $x_r \in \{-1,+1\}$ is the observed sign and R_i contains distinct acquisition events for instance i.

The oracle uses generated observation values rather than model-interpreted values. A secondary posterior computed from interpreted root values helps distinguish perception errors from double counting. No independence-based oracle is applied to the related-view condition.

Let $A_i=1$ denote an actual approval and $A_i=0$ denote deferral or abstention. The primary decision metrics are:

- **Premature approval rate:** the fraction of oracle-ineligible cases approved,

$$\mathrm{PAR} = \frac{\sum_i \mathbf{1}[p_i^* < \tau]A_i}{\sum_i \mathbf{1}[p_i^* < \tau]}.$$

- **Corroboration retention:** the fraction of oracle-eligible independent-observation cases approved. If $\mathscr{I}$ denotes that condition,

$$\mathrm{CR} = \frac{\sum_{i\in\mathscr{I}} \mathbf{1}[p_i^* \geq \tau]A_i}{\sum_{i\in\mathscr{I}} \mathbf{1}[p_i^* \geq \tau]}.$$

- **Incorrect approval rate:** the fraction of negative-state cases approved,

$$\mathrm{IAR} = \frac{\sum_i \mathbf{1}[Y_i = 0] A_i}{\sum_i \mathbf{1}[Y_i = 0]}.$$

All rates include their denominators; a rate with no eligible instances is reported as undefined. Premature approval measures disagreement with the reference policy, while incorrect approval measures an error against the realized hidden state.

Additional outcomes include signed confidence error $\hat{p}_i - p_i^*$, mean absolute confidence error, Brier score $(\hat{p}_i - Y_i)^2$, and paired confidence changes between single-observation and repeated-evidence conditions.

For one positive observation presented $m > 1$ times, we adapt the effective-reading and copy-weight measurements of Gao et al. [2]:

$$\hat{e}_i = \frac{\mathrm{logit}(\hat{p}_i)}{\log(q/(1-q))}, \qquad w_m = \frac{\bar{\hat{e}}_m - 1}{m-1}.$$

A copy weight of zero indicates no additional evidence assigned to the copies under the calibrated reference model. The single-report offset is reported alongside this quantity to expose perception or calibration errors. Probabilities are clipped to $[10^{-6}, 1-10^{-6}]$ only for logit calculations.

Abstention, malformed-output frequency, matching errors, token usage, metadata size, reconciliation calls, and end-to-end latency quantify practical costs. Confidence metrics use valid probability outputs and report their coverage; decision metrics retain all attempted cases.

Paired method differences receive 95% bootstrap intervals resampling whole base instances, keeping their conditions and repeated evaluations together. The analysis reports effect sizes, uncertainty, and denominators.

4. Analytical Results and Expected Outcomes

4.1 Analytical Demonstration of Premature Approval

Consider one positive observation with reliability $q = 0.75$ and prior probability $1/2$. Its likelihood ratio is

$$\frac{q}{1-q} = 3,$$

so its correct posterior is

$$p^* = \frac{3}{1+3} = 0.75.$$

If an aggregator incorrectly treats three presentations of this observation as independent acquisitions, it multiplies the same likelihood ratio three times:

$$p_{\text{duplicate}} = \frac{3^3}{1+3^3} = \frac{27}{28} \approx 0.9643.$$

At threshold $\tau = 0.90$, this changes the decision from defer to approve without adding an acquisition. Three genuinely independent positive observations also yield 0.9643, but legitimately under the controlled independence model.

This result establishes the benchmark's failure mechanism and a sanity check for the oracle and gate. It does not establish how frequently an LLM will exhibit the behavior.

Evidence treatment	Posterior	Decision
One positive acquisition	0.7500	Defer
Three copies counted as independent	0.9643	Approve
Three copies counted once	0.7500	Defer
Three independent positive acquisitions	0.9643	Approve

Table 2: Analytical illustration with $q = 0.75$ and $\tau = 0.90$. These are calculated values, not measured LLM outputs.

4.2 Hypotheses and Interpretation Criteria

The main hypothesis is that trusted-lineage consolidation reduces premature approvals on transformed and cross-modal copies while preserving approvals supported by independent observations.

Exact hashing should identify byte-identical replay but cannot generally link a caption to its source image. Perceptual matching may recognize image transformations, but can merge similar independent acquisitions or miss cross-modal descendants. Whether lineage provides a meaningful advantage over a clear instruction remains an empirical question.

Evidence supporting the hypothesis would consist of lower premature approval rates and confidence error, together with preserved corroboration retention and acceptable overhead. These outcomes must be considered jointly: reducing approvals through blanket abstention is not sufficient.

Comparable performance from a cheaper instruction-based method would narrow the need for explicit lineage. Poor retention would expose information lost during consolidation. A measurable benefit limited to cross-modal cases would support a correspondingly narrower contribution.

No project experiments are reported here. The numerical results in Table 2 are analytical calculations; the hypotheses above remain to be tested.

4.3 Limitations

Trusted lineage is a strong assumption. Forged roots, missing transformations, and dishonest acquisition systems can undermine it. Distinct acquisition identifiers do not establish independence in real environments, where measurements may share causes.

Shared-root reports can improve extraction, so consolidation may lose useful interpretation diversity. The reconciliation stage addresses this partially but introduces additional model error and cost. Synthetic visual cards provide precise controls but have limited ecological validity. One model and 100 base instances support a bounded case study rather than a general claim about all multimodal agent systems.

5. Timeline, Milestones, and Expected Contribution

5.1 Milestones and Acceptance Criteria

The project schedule spans October 1–December 2. Table 3 lists planned work packages, deliverables, and completion criteria. Dates describe the schedule rather than establishing that a milestone is complete.

Table 3: Project schedule from October 1 to December 2.

Period	Work package, deliverable, and completion criterion
Oct. 1–7	**Problem formulation and literature review.** Review epistemic Sybil resistance, evidence dependence, and provenance. Deliver a focused problem statement, research objectives, and contribution boundary.
Oct. 8–14	**Experimental design.** Specify the controlled task, transformation conditions, baselines, and metrics. Deliver a design separating report multiplicity from acquisition multiplicity.
Oct. 15–21	**Benchmark specification and report preparation.** Finalize observation-generation rules, oracle equations, provenance fields, and implementation scope. Deliver the project plan with explicit assumptions and limitations.
Oct. 22–28	**Benchmark and provenance.** Generate seeded observations, oracle labels, and lineage. Verify roots survive replay, cropping, compression, and captioning.
Oct. 29–Nov. 4	**Pipeline and pilot.** Integrate workers, aggregation, and the action gate. Pass lineage and parsing checks across core conditions; freeze prompts, thresholds, model version, matching settings, and spending cap.
Nov. 5–11	**Baseline evaluation.** Run paired inputs across baseline methods. Deliver cached outputs and audits of malformed responses and information-losing transformations.
Nov. 12–18	**Defense and ablations.** Run consolidation comparisons with and without ancestry. Include distinct-observation controls and unknown-lineage cases.
Nov. 19–25	**Analysis and validation.** Deliver confidence, decision-error, uncertainty, and overhead tables. Categorize failures and verify retention of legitimate approvals.
Nov. 26–Dec. 2	**Reproducible release and demonstration.** Package inputs, prompts, lineage, outputs, and scripts. Complete the report and demonstration; regenerate all result tables from saved outputs.

5.2 Responsibilities and Scope Control

One team member owns observation generation, oracle calculations, and lineage propagation. The other owns model integration, baseline matching, and evaluation scripts. Both review prompts, audit failures, interpret results, and prepare the final report.

Full experimental runs depend on passing pilot lineage, parsing, and decision-gate checks. If cost or integration delays arise, single-observation, exact-replay, transformed-copy, cross-modal-replay, and independent-observation conditions take priority. Related views, a second model, multi-root summaries, and prompt-injection stress tests follow only after the core comparison.

5.3 Expected Contribution

The project is expected to produce three technical artifacts.

Controlled multimodal benchmark. The benchmark will separate repeated representations from distinct acquisitions while holding the underlying task fixed. Its controlled conditions will make confidence changes and decision errors attributable to defined evidence structures.

Minimal provenance interface and consolidation prototype. The root-and-parent interface will preserve acquisition identity across transformations and agent reports. The prototype will

consolidate repeated representations while retaining uncertainty and conflicting interpretations.

Comparative empirical evaluation. The evaluation will quantify premature approvals, incorrect approvals, corroboration retention, calibration, and overhead relative to instructions and content matching.

Together, these artifacts will identify where lineage improves on simpler methods, where it does not justify its cost, and where consolidation loses useful interpretation diversity. The conclusions will remain bounded by the controlled generator and trusted-lineage assumptions.

6. AI Usage

ChatGPT assisted with literature discovery, report organization, drafting, analytical illustration, and LaTeX preparation. No project experiments were executed for this report. The numerical illustration is derived from the stated Bayesian model and is not presented as measured system performance.

References

[1] M. Bara, "Epistemic Sybil Resistance: Multiplying AI Agents Without Multiplying Evidence," *arXiv preprint arXiv:2609.01873*, 2026. [Online]. Available: https://arxiv.org/abs/2609.01873.

[2] J. Gao, R. Li, T. Yu, L. Ren, B. Chen, and Z. Wang, "Copies or Sources? Measuring How LLM Aggregators Count Restated Evidence in Multi-Agent Systems," *arXiv preprint arXiv:2610.06192*, 2026. [Online]. Available: https://arxiv.org/abs/2610.06192.

[3] J. Cheney, A. Ahmed, and U. A. Acar, "Provenance as dependency analysis," *Mathematical Structures in Computer Science*, vol. 21, no. 6, pp. 1301–1337, 2011. doi: 10.1017/S0960129511000211. [Online]. Available: https://doi.org/10.1017/S0960129511000211.

Preliminary implementation results

ADDENDUM | October 30, 2026 | Software diagnostics and live-pilot status

Scope update. This dated addendum supplements the original report. Its statements in Sections 4.2 and 6 that no experiments were executed describe the October 28 report. Since then, offline software diagnostics and an incomplete live API attempt were executed. No successful real-vision comparison is reported here.

A. Verified offline transformation diagnostics

Ten balanced synthetic base instances, six paired conditions, four programmed methods: 240 evaluations. Sensor reliability is 0.75, prior 0.5, and approval threshold 0.90. Workers use fixture observations, not image perception. These results verify implementation behavior; they are not measured LLM effectiveness.

Mean absolute oracle error	Naive	Hash	Lineage	Perceptual
Single raster	0.0000	0.0000	0.0000	0.0000
Exact replay	0.2143	0.0000	0.0000	0.0000
Resize + JPEG	0.2143	0.2143	0.0000	0.0000
Crop replay	0.2143	0.2143	0.0000	0.1500
Synthetic cross-modal	0.2143	0.2143	0.0000	0.2143
Independent acquisitions	0.0000	0.2357	0.0000	0.2357

Each cell averages 10 valid software outputs. Naive replay prematurely approves 4/10 oracle-ineligible cases. Hashing also approves 4/10 in each transformed/cross-modal replay condition because bytes differ. Trusted lineage has zero oracle error in these fixtures. In the independent condition, naive and lineage retain 2/2 oracle-supported approvals; hash and perceptual matching retain 0/2 after merging equal-looking independent observations.

B. Verification and interpretation

The original pilot and transformation commands run. Artifact manifests verify; the original pilot outputs are unchanged. Regression tests cover ancestry, actual-byte hashes, extraction boundaries, cache replay/integrity, quota stopping, and failure denominators. Transformed glyphs are readable in the contact sheet, and the known-geometry signal checks pass.

Calibration roots are separate from evaluation, but appearance templates overlap. The perceptual matcher is image-only and crop-sensitive. Acquisition independence is stipulated by the synthetic generator; root IDs alone do not establish independence in real settings. These diagnostic outcomes must not be used as evidence that an LLM defense works.

Source: artifacts/transformations/summary.json and reviewed provenance/signal checks. Values are regenerated from saved outputs by scripts/build_preliminary_report.py.

Real-vision pilot: incomplete

GEMINI 3.8 FLASH | Billing-disabled free tier | $0 cap | No paid fallback

C. Implemented model pipeline

The opt-in adapter sends actual PNG/JPEG bytes to Gemini and validates structured observations. A separate image-to-record call can create the cross-modal derivative; trusted code preserves its parent and acquisition root. Naive, instruction-only, and trusted-lineage methods share cached worker reports and the same aggregator model. Non-null probabilities must come from model responses with valid evidence citations. Ground truth and oracle values remain evaluator-only.

D. Observed execution outcome

Two-instance stage	Observed outcome
Scheduled method-condition rows	36
Attempted aggregation evaluations	2
Valid aggregation outputs	0
Vision extraction calls / resolved	1 / 0
Completed base-instance comparisons	0
Ten-instance live stage	Not started
Run status	Stopped: HTTP 503 (high demand)

The first image request timed out after 90 seconds and was not resent. An authenticated read-only model lookup succeeded. After manual review, remaining requests continued: one aggregator response failed citation validation, and the next received HTTP 503 indicating high demand. Execution stopped. Raw outcomes, model settings, timing, usage when available, and failure details are preserved. A separate sandbox connectivity failure is excluded from these live-stage counts.

No empirical vision accuracy, valid aggregation comparison, or efficacy interval is available. The 36 scheduled rows remain explicit: two attempted failures and 34 not-attempted rows. Null probabilities and failed requests are not converted into successful abstentions. A zero approval count in this incomplete run is not evidence of safety or improved calibration.

E. Next experimental step

When free-tier service is available, review the stopped run and plan an explicitly documented continuation or new frozen replicate. Preserve timeout outcomes rather than silently retrying or dropping them. Complete and inspect the two-instance stage before ten. Only then report paired whole-base bootstrap intervals, extraction accuracy, confidence inflation, Brier score, premature/incorrect approvals, retention, and coverage. Small synthetic samples cannot establish natural-image performance or general LLM defense effectiveness.

AI usage update: Codex assisted with implementation, tests, API execution, diagnostic analysis, and this addendum. The original report is retained unchanged; no successful live result is implied by its prospective methodology.

Evidence: artifacts/real-vision-gemini/{run_status.json,results.jsonl,extraction_summary.json,cache/}; protocol: docs/real-vision-pilot.md. Official model and free-tier references: ai.google.dev/gemini-api/docs/models/gemini-3.8-flash and ai.google.dev/gemini-api/docs/pricing (checked October 30, 2026).